



Genomics of Gene Regulation: Concepts and **Techniques**

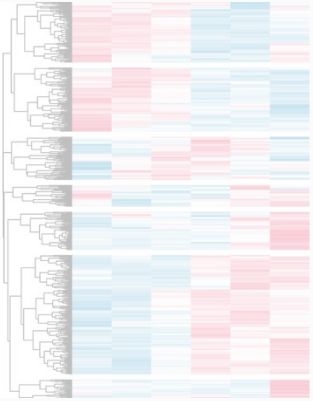
Single Cell technologies
Enhancer analysis

Ana Azambuja

Senior Research Associate
Pathology Department - Boston Children's Hospital

Topics to be covered - Techniques

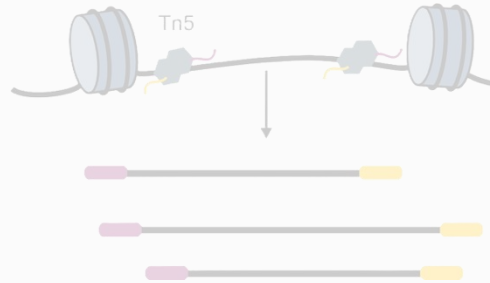
GENE EXPRESSION



RNAseq

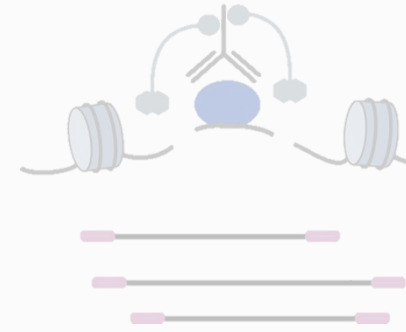
Spatial transcriptomics

EPIGENOMIC REGULATION



ATACseq

DNA-PROTEIN BINDING



ChIPseq
CUT&RUN
CUT&TAG

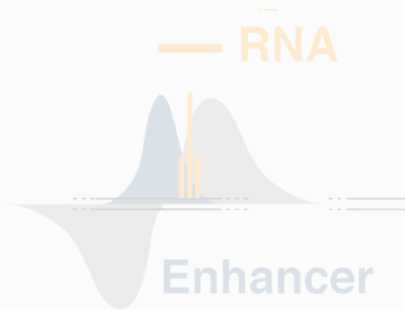
GENOME ORGANIZATION

3D genome organization - HiC



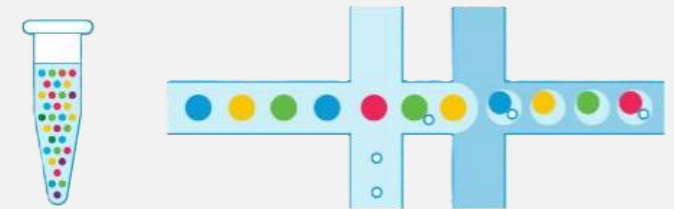
ENHANCER ACTIVITY

eRNA mapping

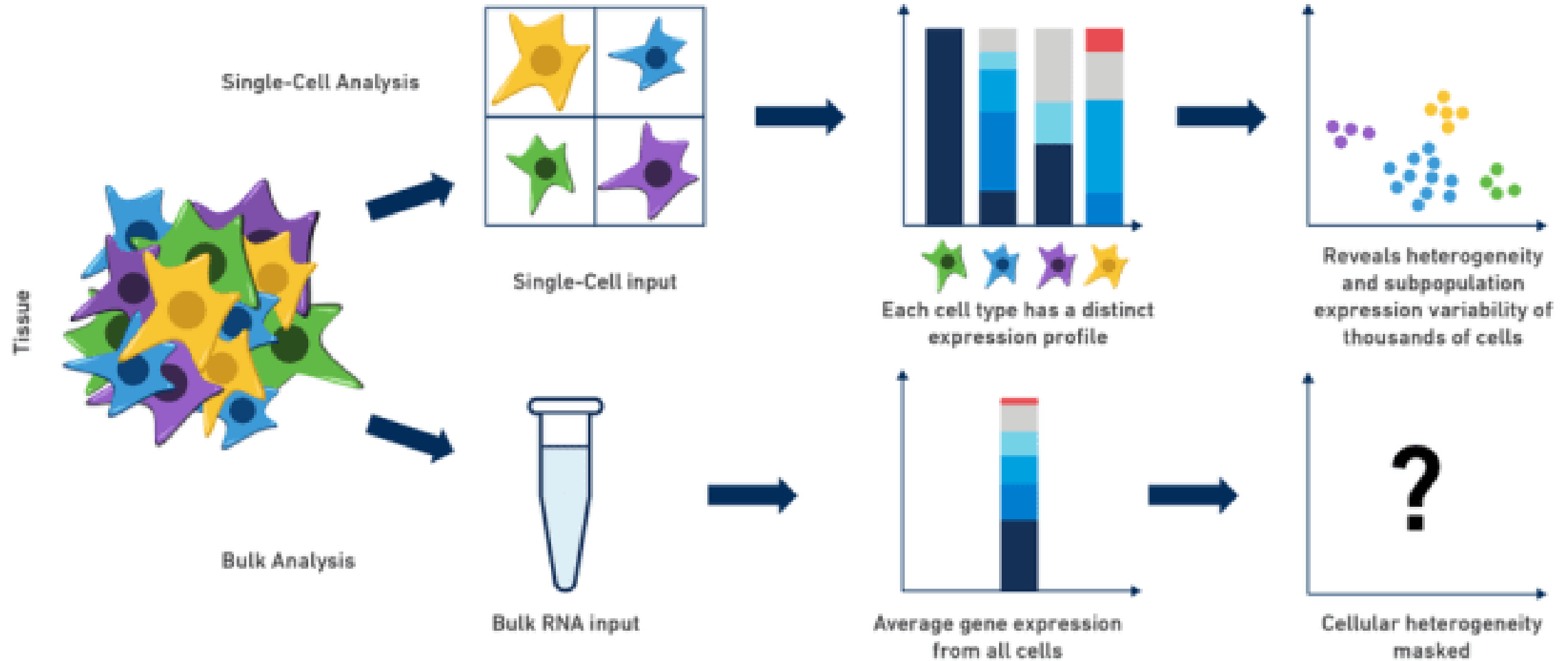


SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome



What about cellular heterogeneity?



Single-cell Technologies evolution

Manual



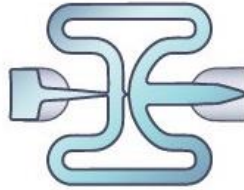
Tang *et al.* 2009¹⁸

Multiplexing



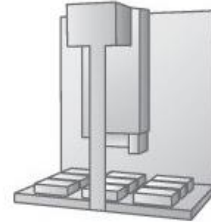
Islam *et al.* 2011²⁴

Integrated fluidic circuits



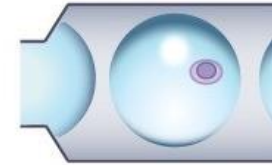
Brennecke *et al.* 2013⁶⁴

Liquid-handling robotics



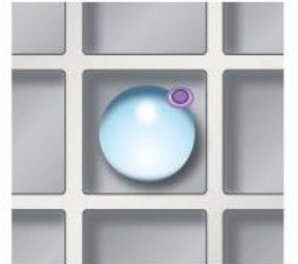
Jaitin *et al.* 2014³³

Nanodroplets



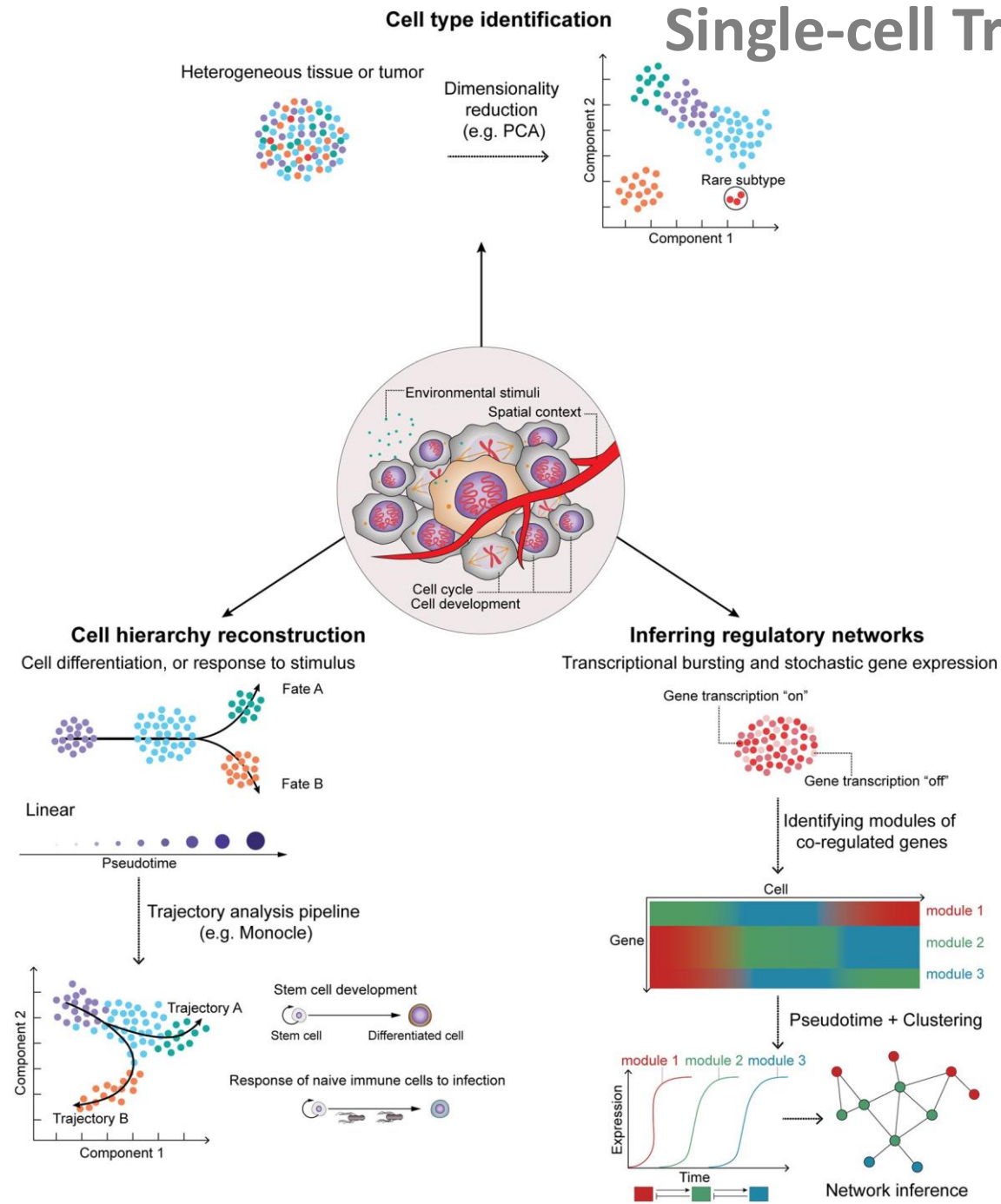
Klein *et al.* 2015³⁴
Macosko *et al.* 2015⁴⁰

Picowells



Bose *et al.* 2015⁴³

Single-cell Transcriptome – Applications



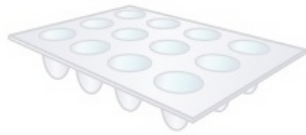
Single-cell Technologies evolution

Manual



Tang *et al.* 2009¹⁸

Multiplexing



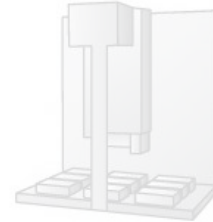
Islam *et al.* 2011²⁴

Integrated fluidic circuits



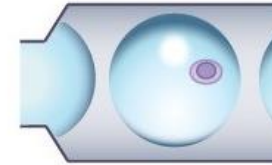
Brennecke *et al.* 2013⁶⁴

Liquid-handling robotics



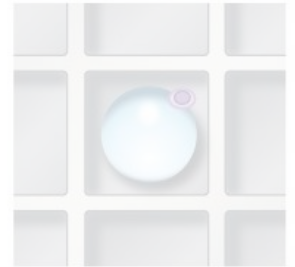
Jaitin *et al.* 2014³³

Nanodroplets



Klein *et al.* 2015³⁴
Macosko *et al.* 2015⁴⁰

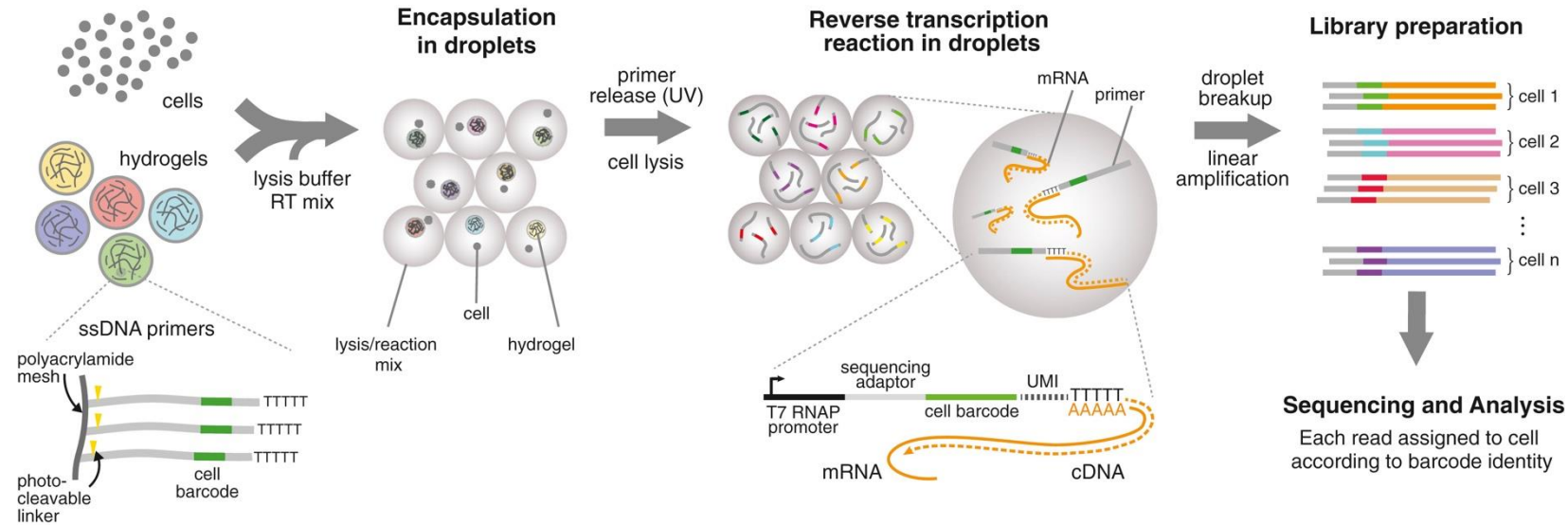
Picowells



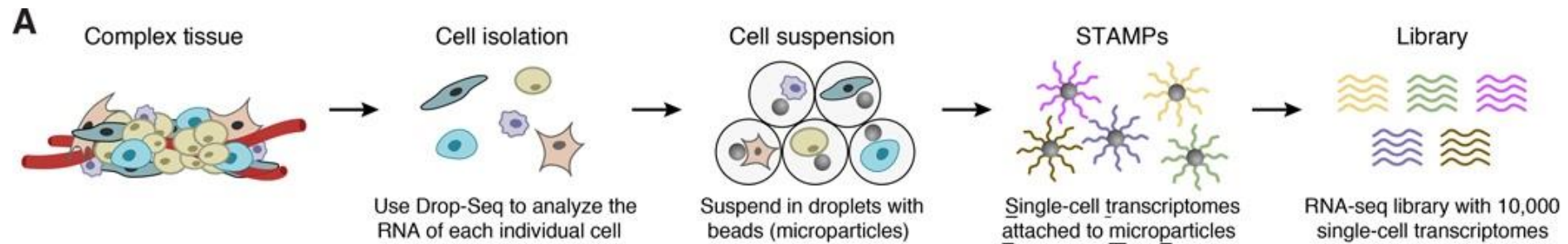
Bose *et al.* 2015⁴³

The advent of microfluidics + nanodroplets

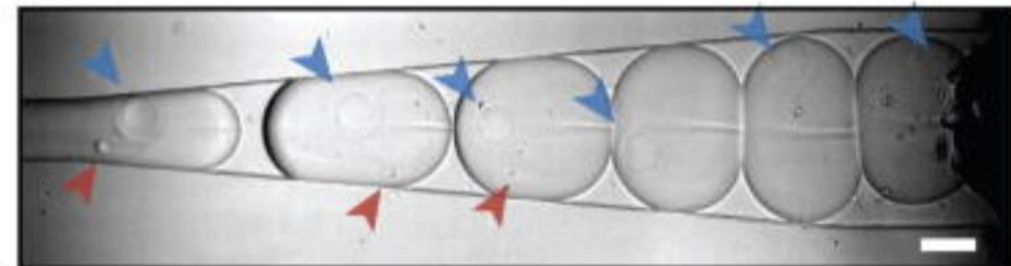
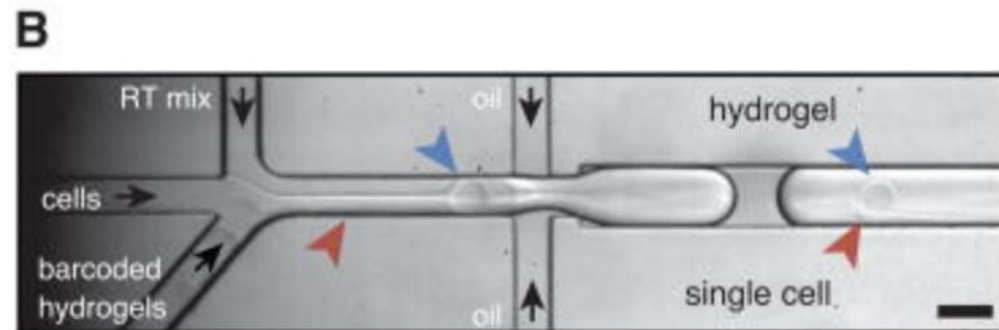
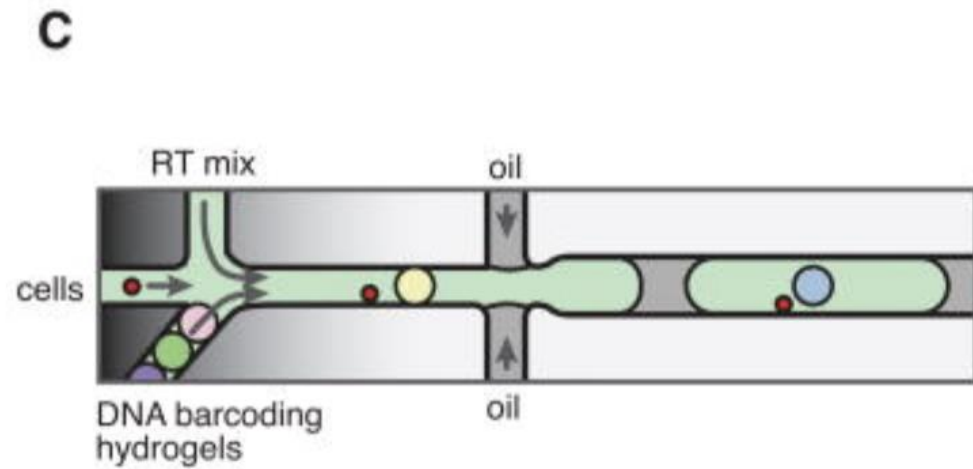
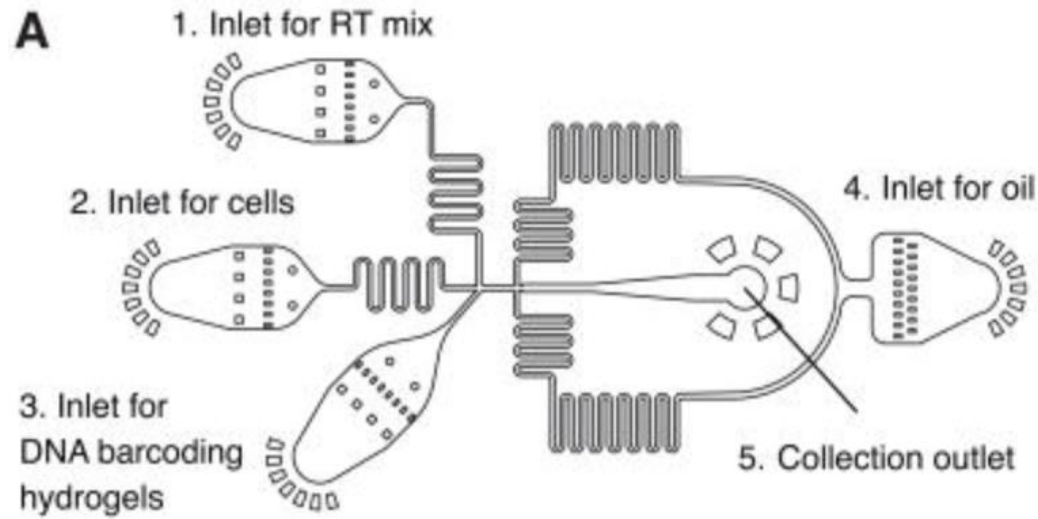
inDrop (indexing Droplets)



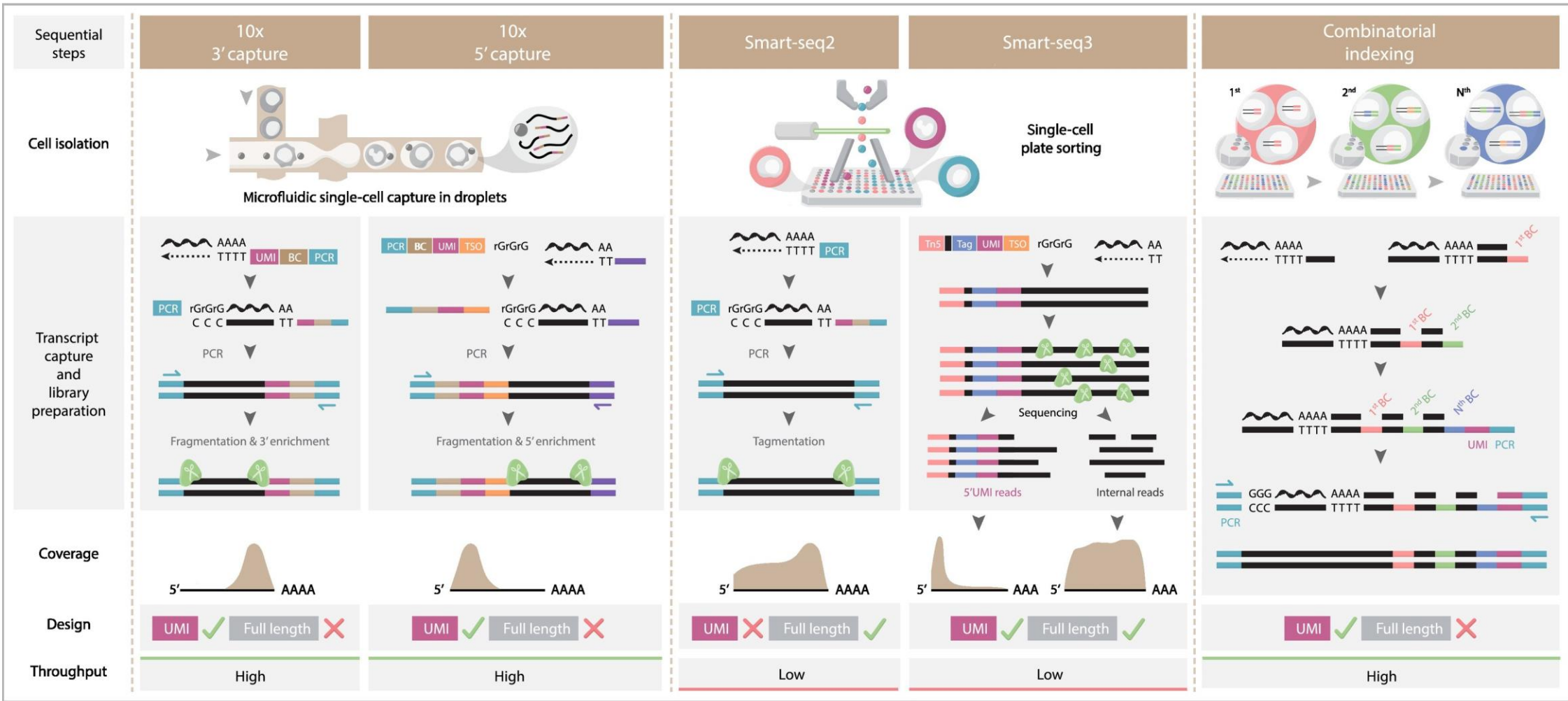
Drop-seq



The advent of microfluidics + nanodroplets

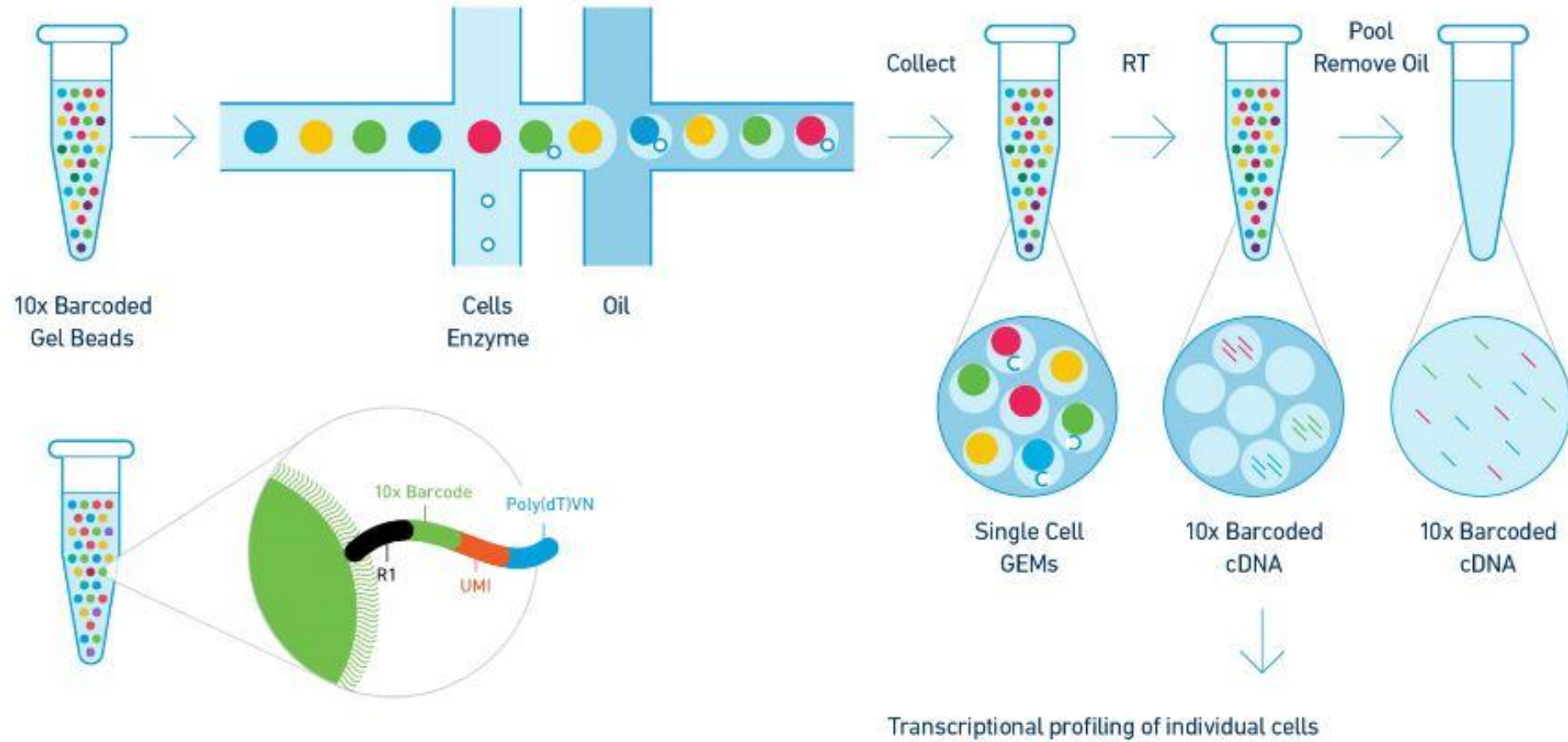


The advent of microfluidics + nanodroplets



Trends in Genetics

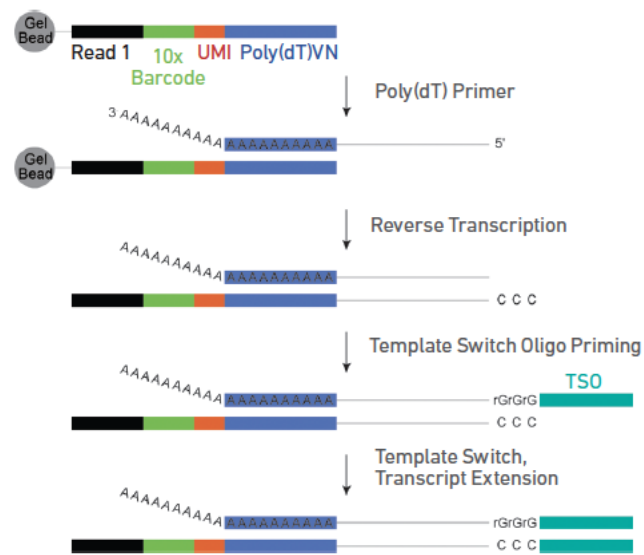
Single-cell Transcriptome – 10x Genomics



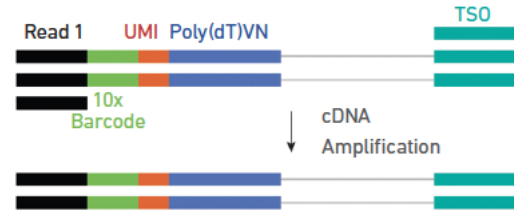
Single-cell Transcriptome – Protocol overview

Inside Gel Beads-in-emulsion (GEMs)

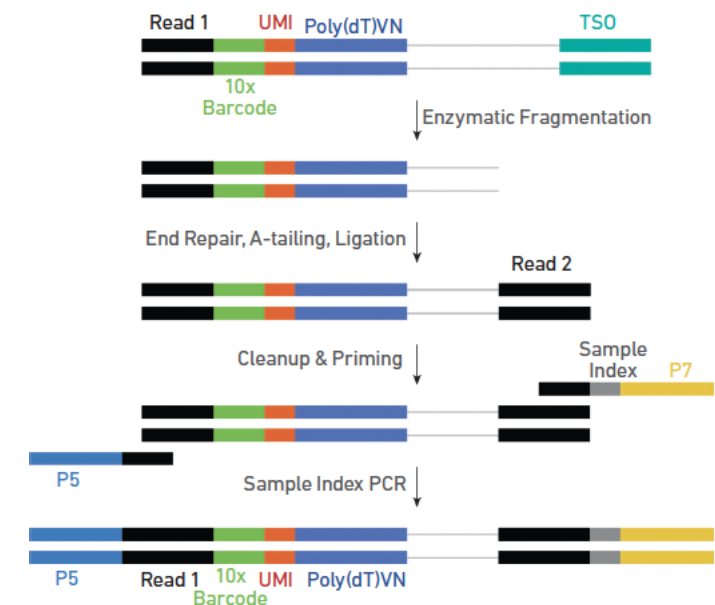
Transcript capture and BC



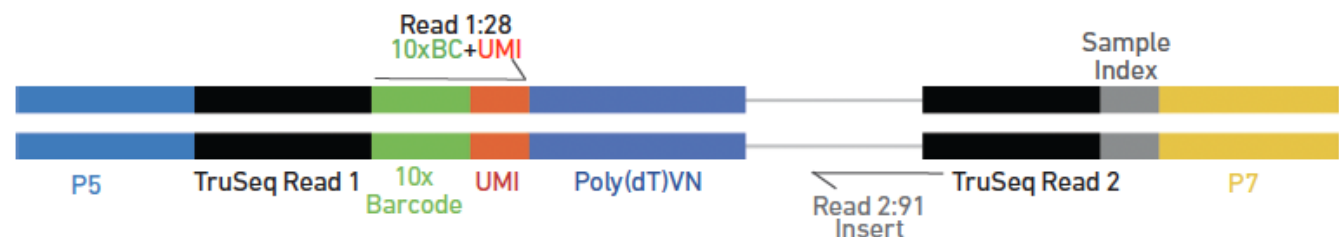
Pooled cDNA amplification



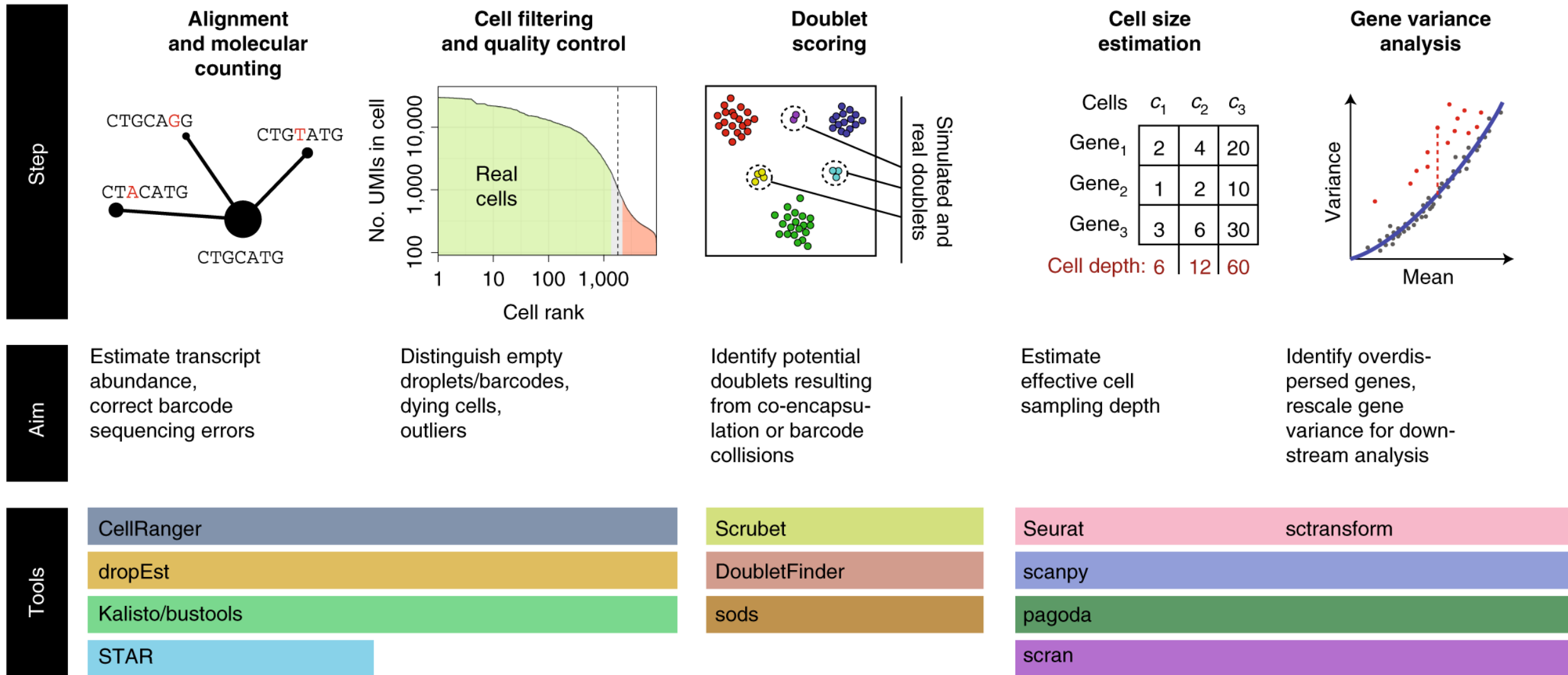
Pooled amplified cDNA processed in bulk



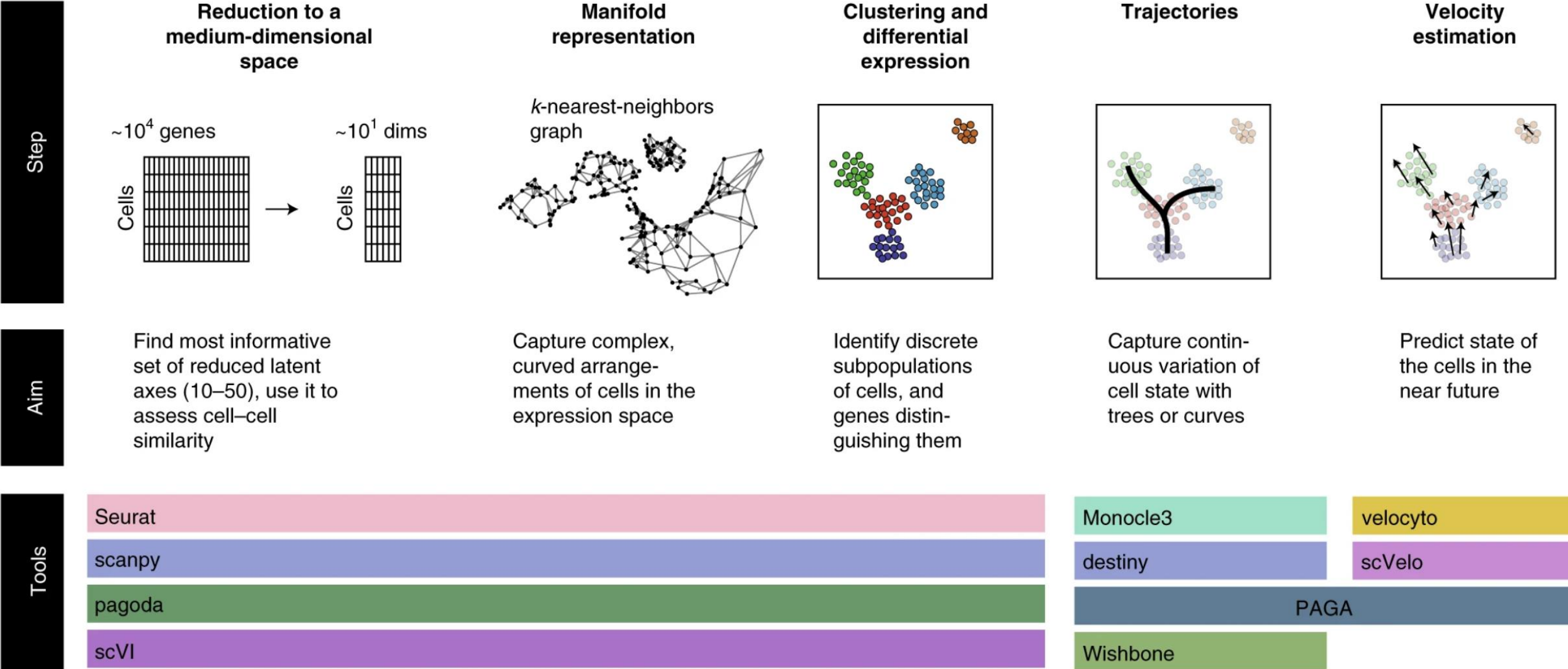
Single Cell 3' Gene Expression Library



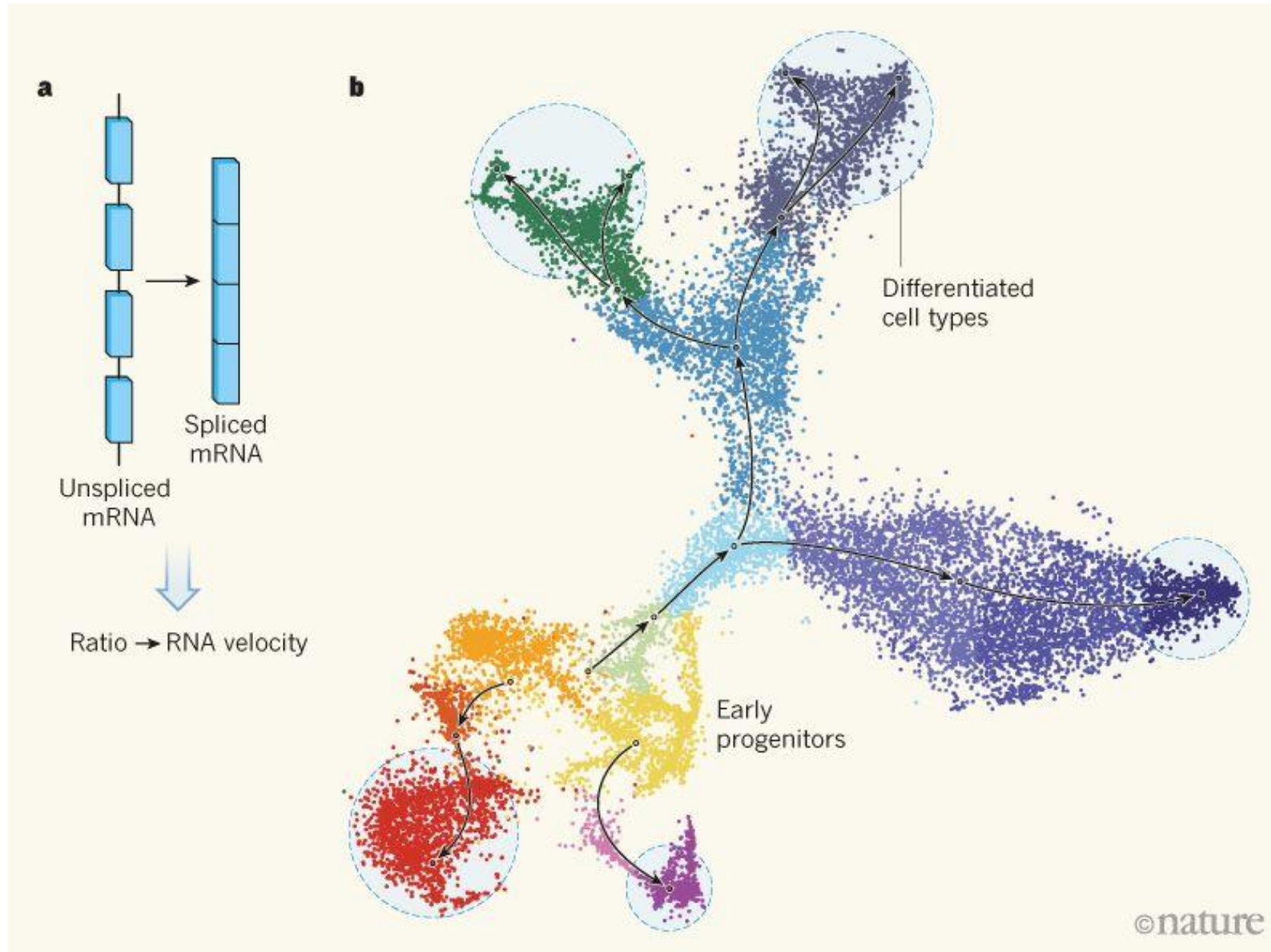
Key preprocessing steps in single-cell RNA-seq analysis



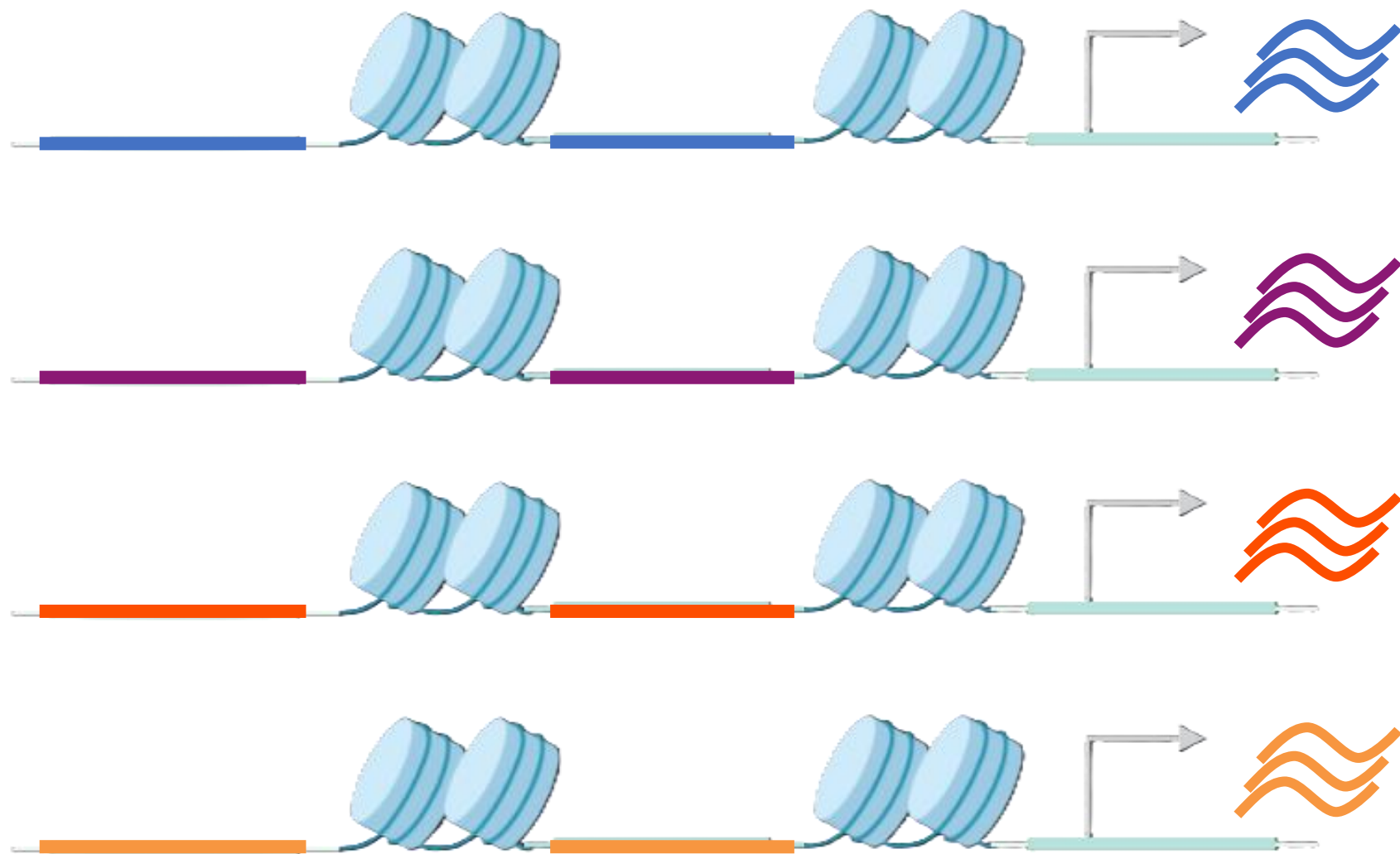
Key analysis steps in single-cell RNA-seq



Measuring the expression dynamics

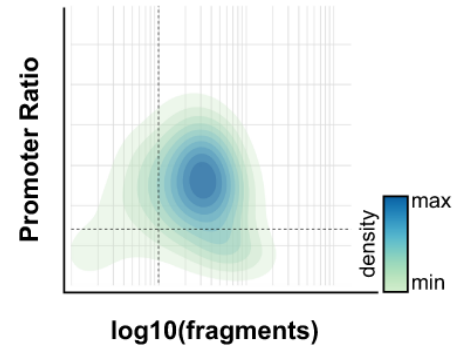


DNA accessibility on a single-cell level

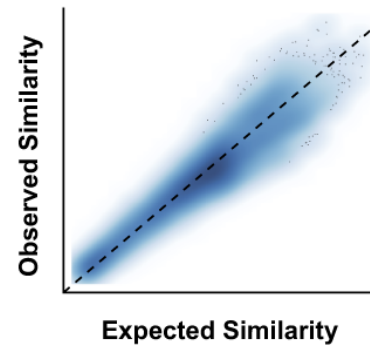


Single-cell ATACseq – Applications

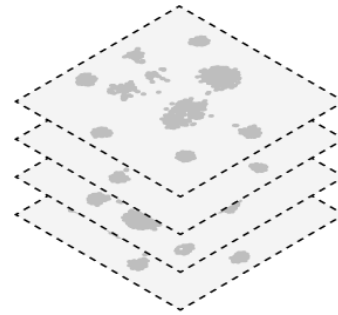
Quality Control Cell Selection



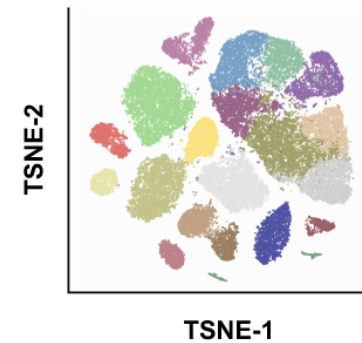
Normalization



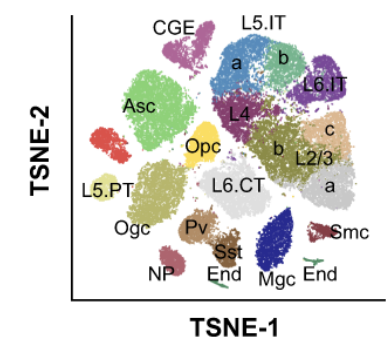
Dimensionality Reduction



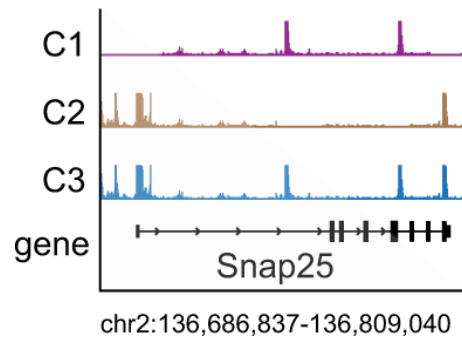
Consensus Clustering



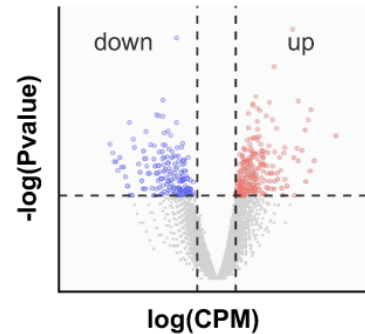
Cluster Annotation



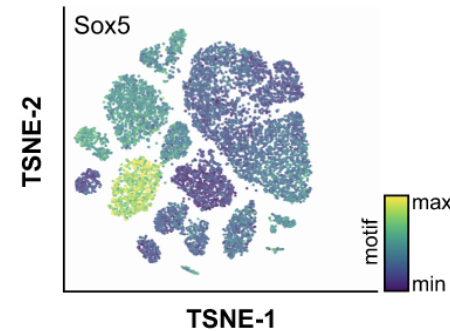
Peak Calling



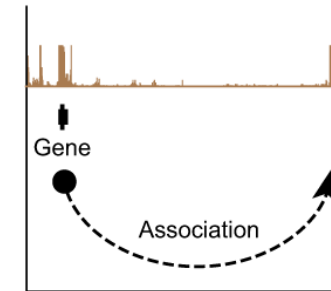
Finding Variable Peaks



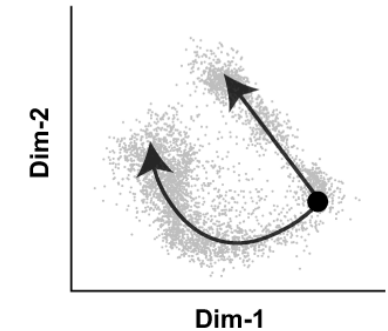
Motif Analysis



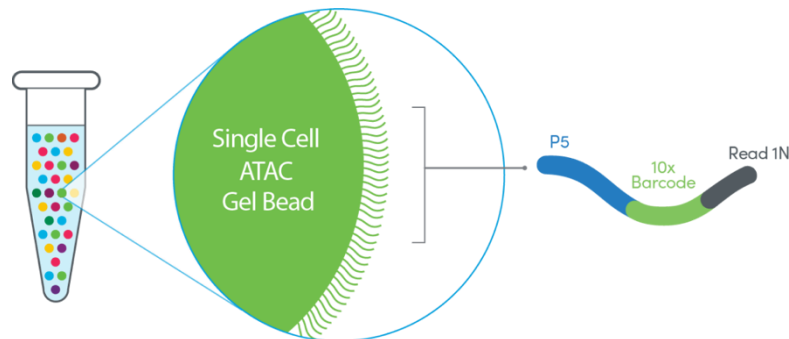
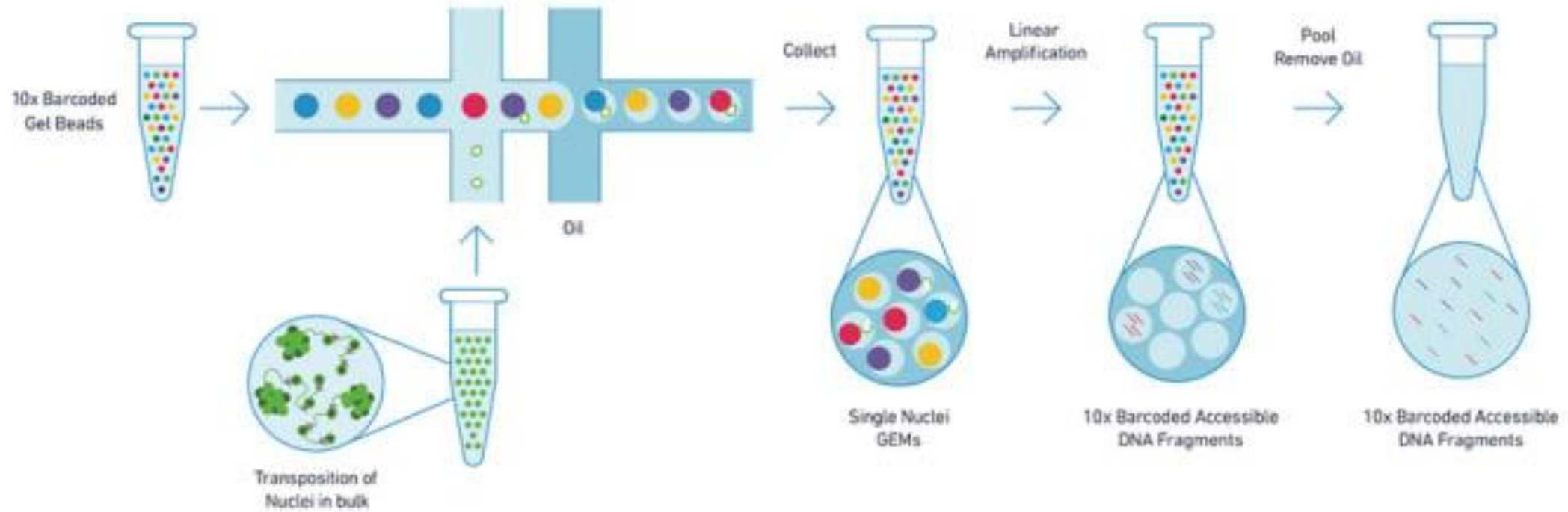
Interaction Prediction



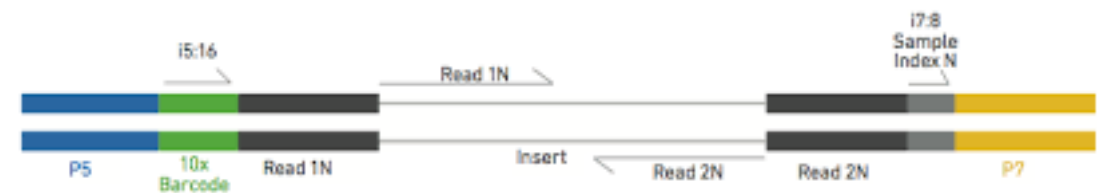
Pseudotime Analysis



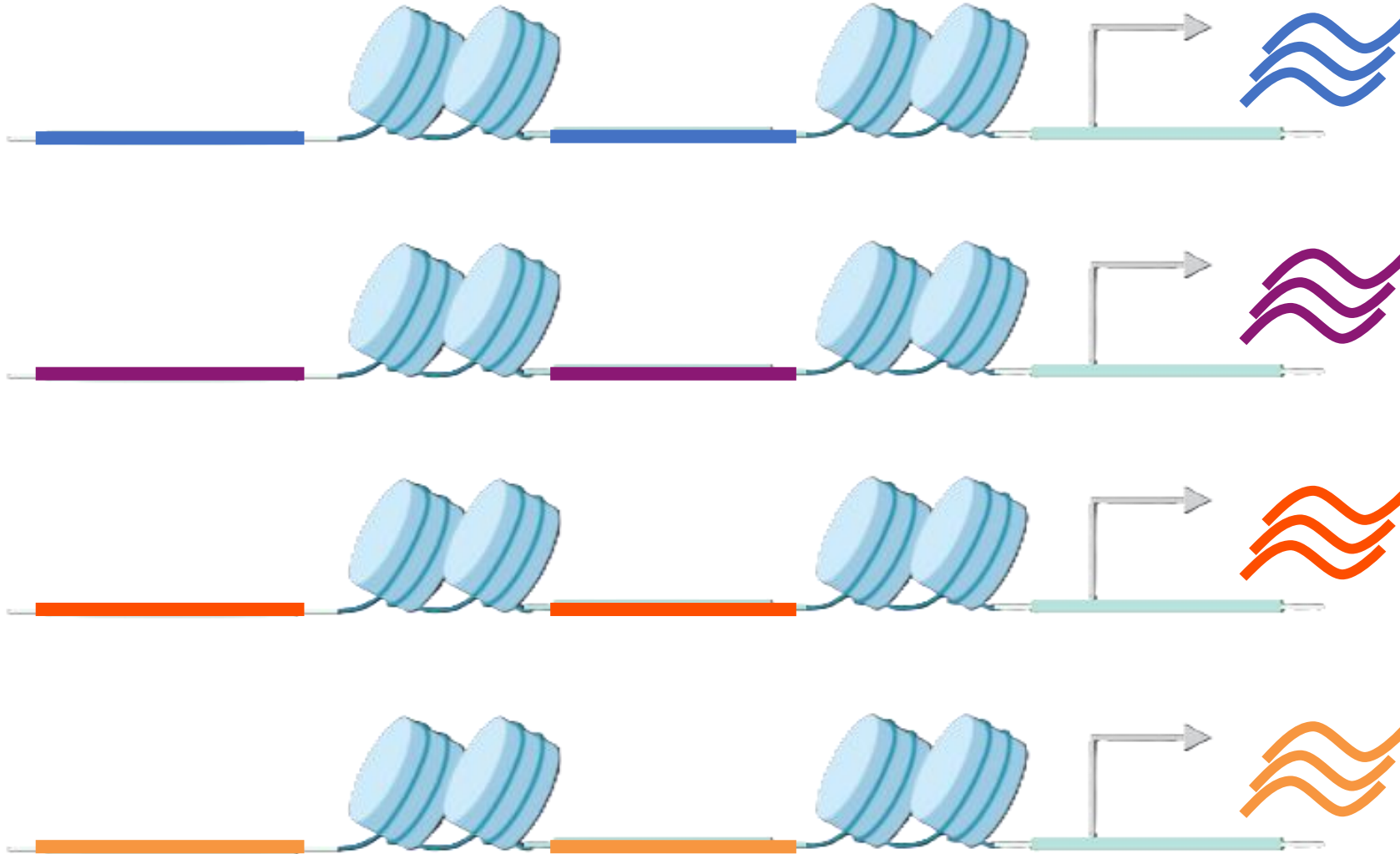
Single-cell ATAC – Protocol overview



Chromium Single Cell ATAC Library



DNA accessibility AND gene expression on a single-cell level



nature methods



METHOD OF THE YEAR 2019

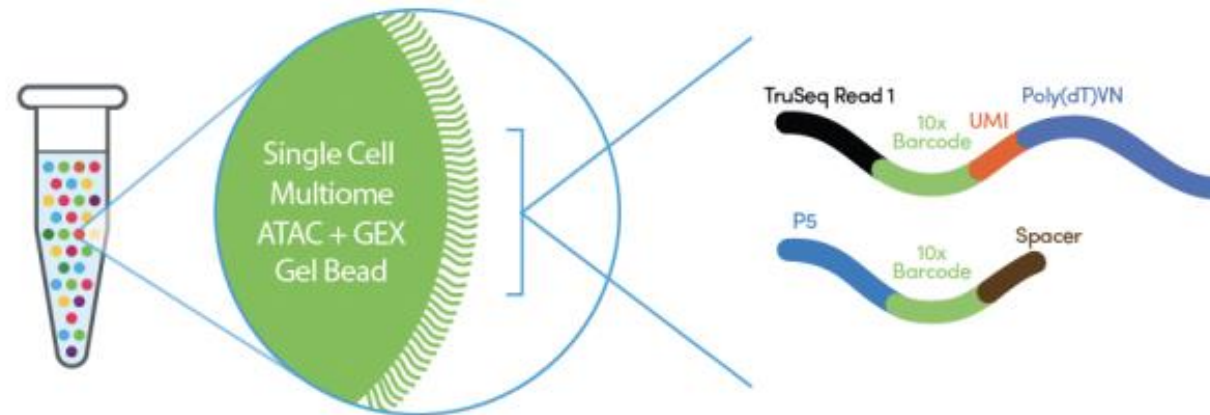
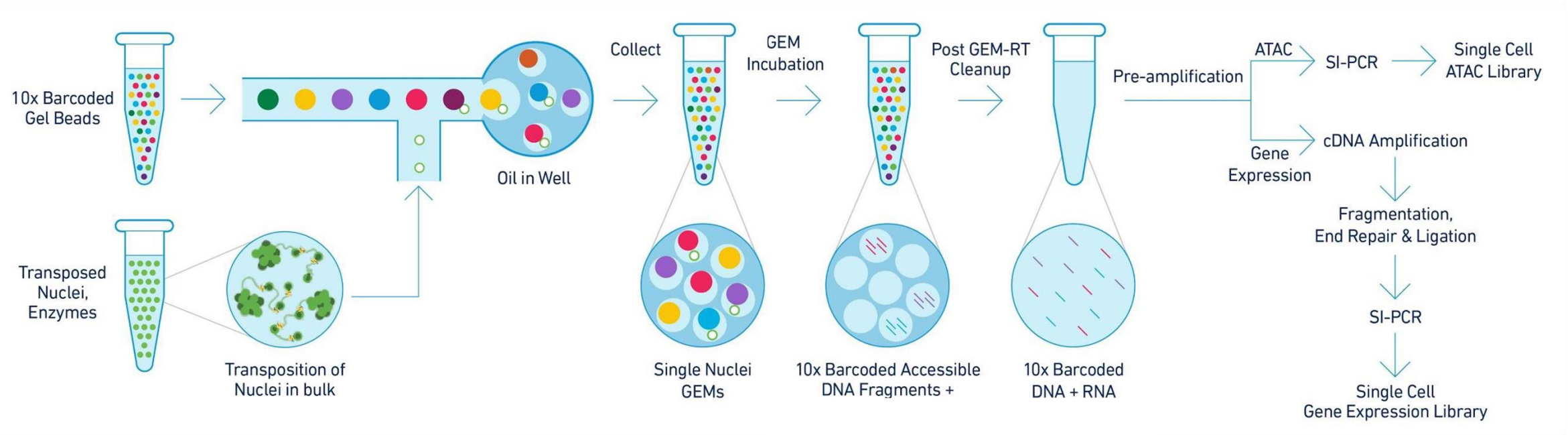
Localization microscopy twice as precise

A cryo-EM-based structural proteomics approach

Time-resolved crystallography at the European XFEL

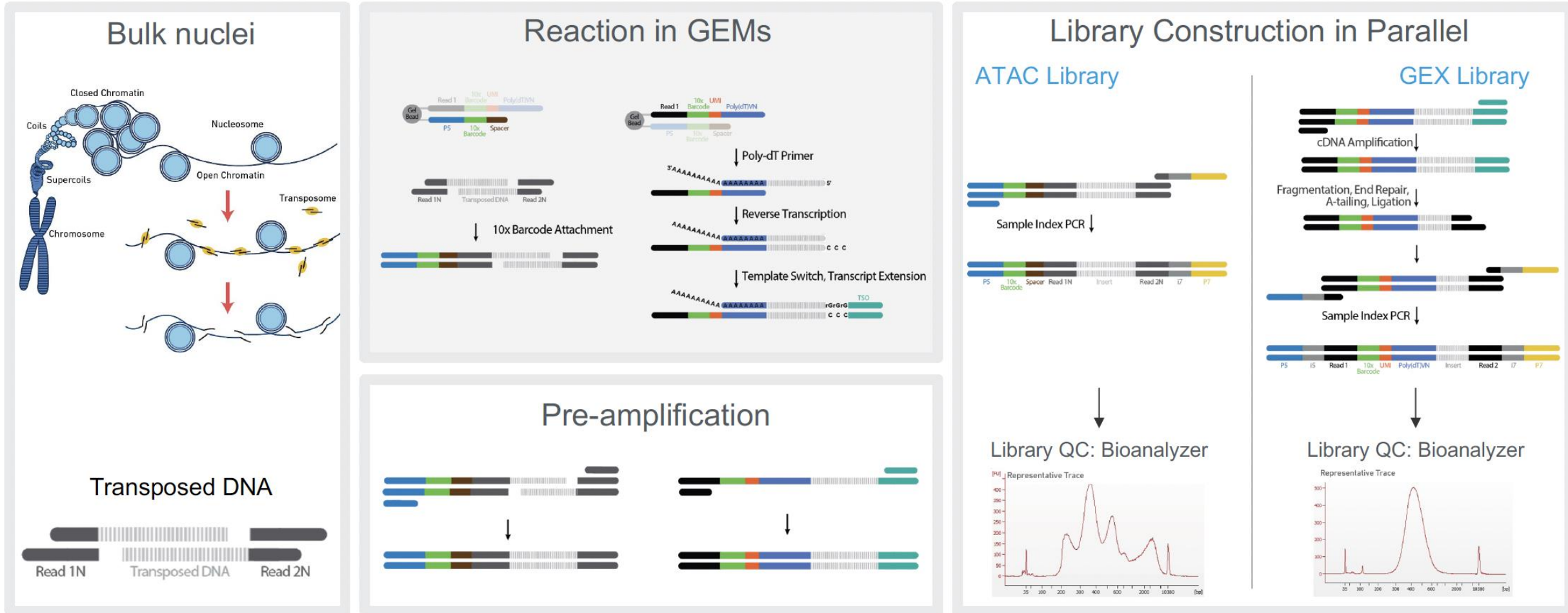
Magnetic resonance at high speed

Single-cell Multiome – Protocol overview



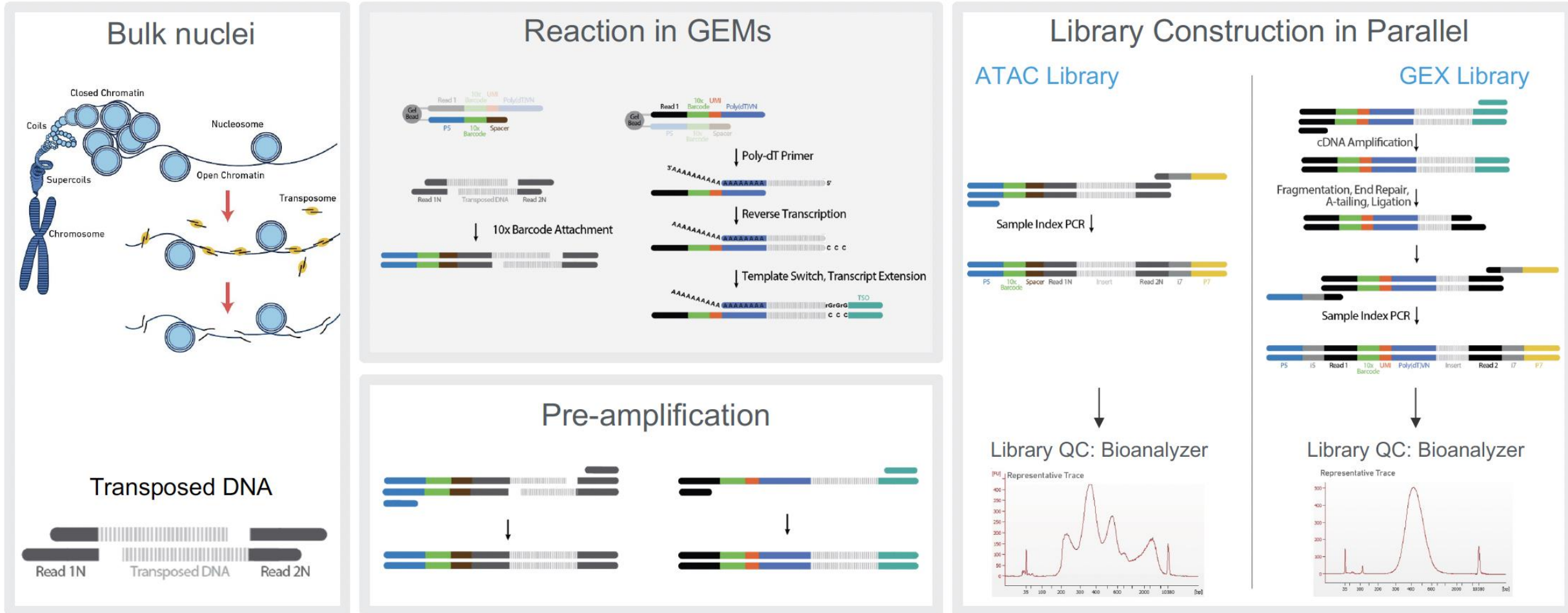
Single-cell Multiome – Protocol overview

Assay scheme

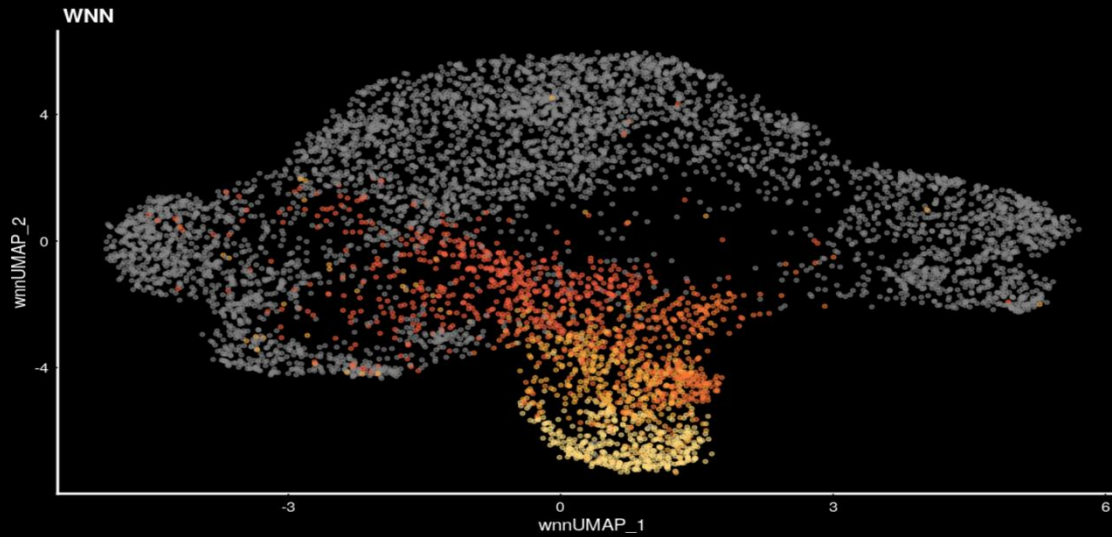


Single-cell Multiome – Protocol overview

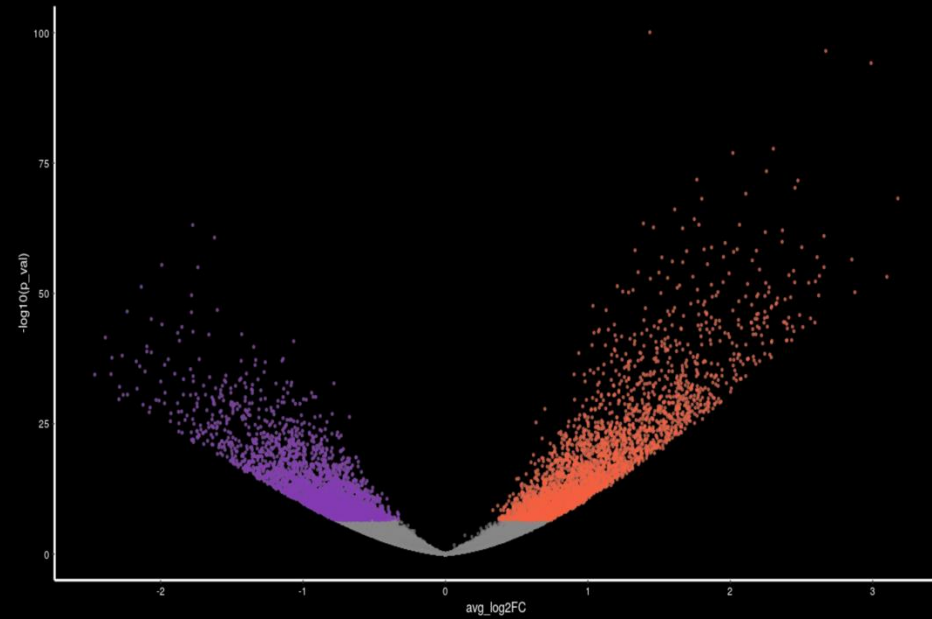
Assay scheme



Mapping chromatin regulated enhancers

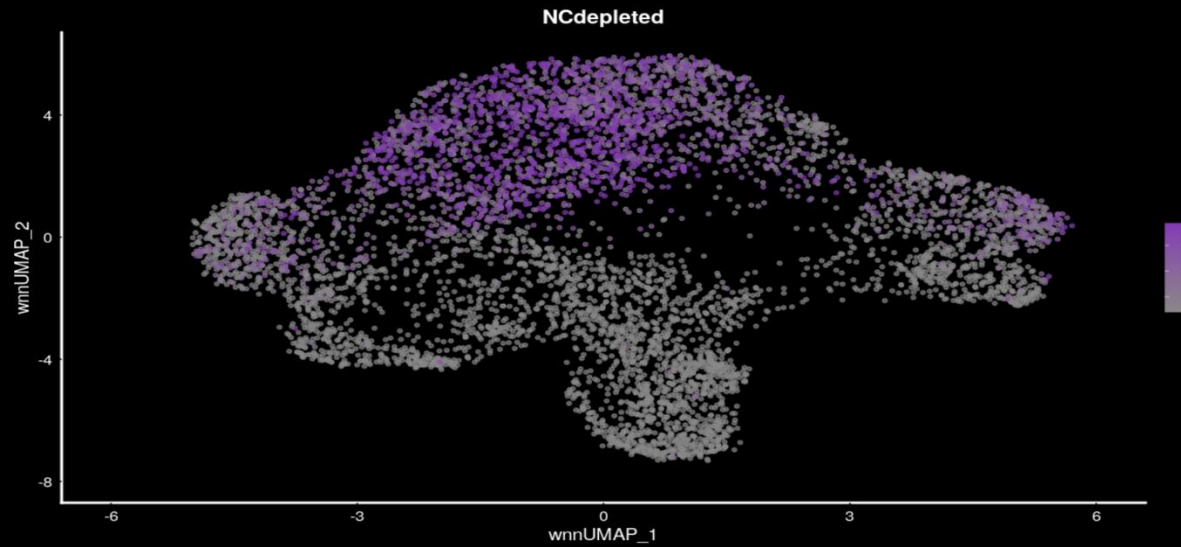


→
differential
accessibility

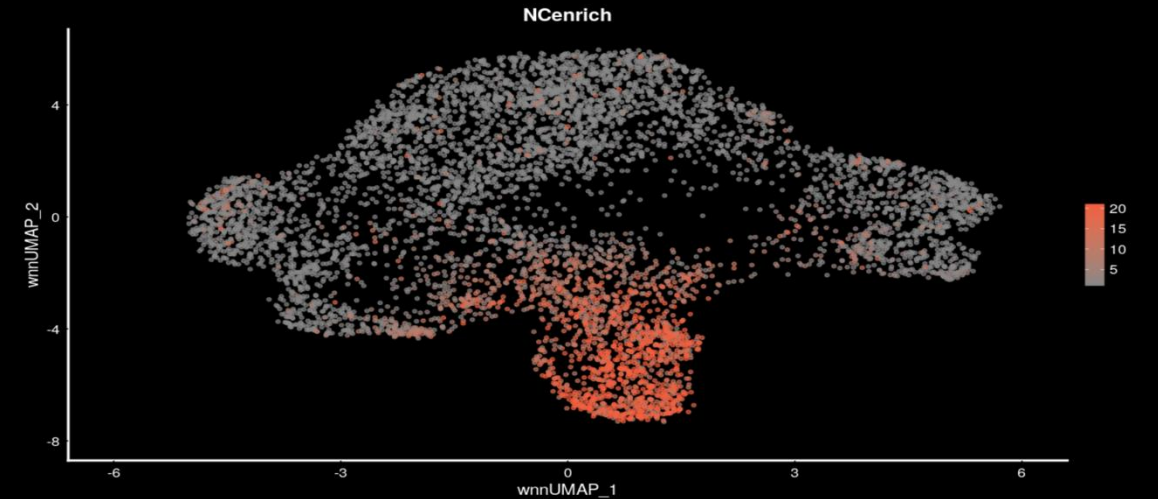


↓
Motif enrichment analysis

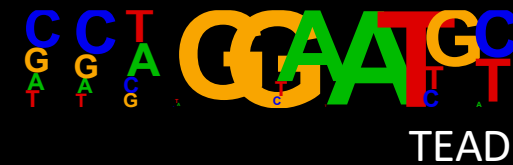
Mapping chromatin regulated enhancers



NF STARR+



NF STARR+



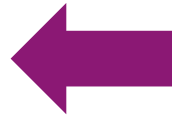
Cis-regulatory features and how to identify them

DNase-seq

DNase I hypersensitive sites sequencing

ATACseq

Assay for Transposase-Accessible Chromatin

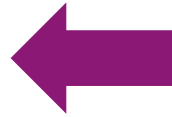


ChIPseq

Chromatin Immunoprecipitation

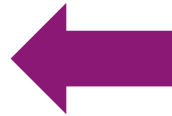
CUT&RUN

Cleavage Under Targets & Release Using Nuclease



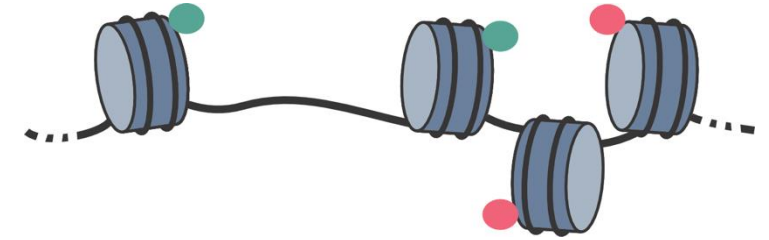
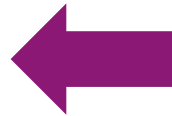
PROseq / GROseq

"nascent" RNA sequencing



HiC

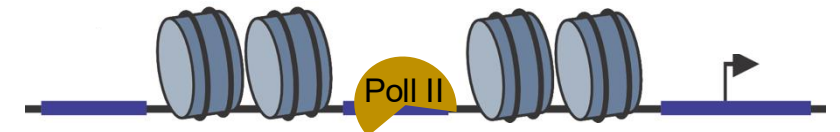
map chromatin contacts genome-wide



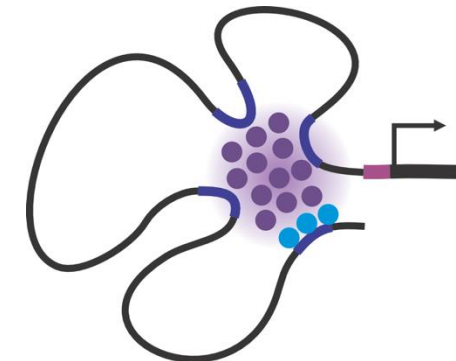
DNA accessibility and active histone marks



Tissue specific transcription factor binding

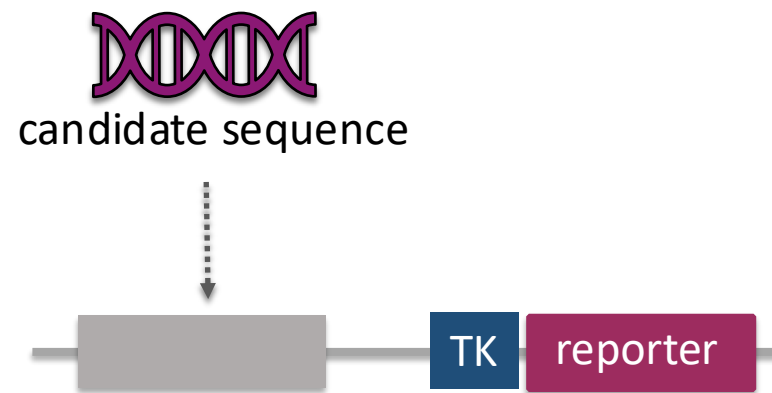


Enhancer activation and transcription

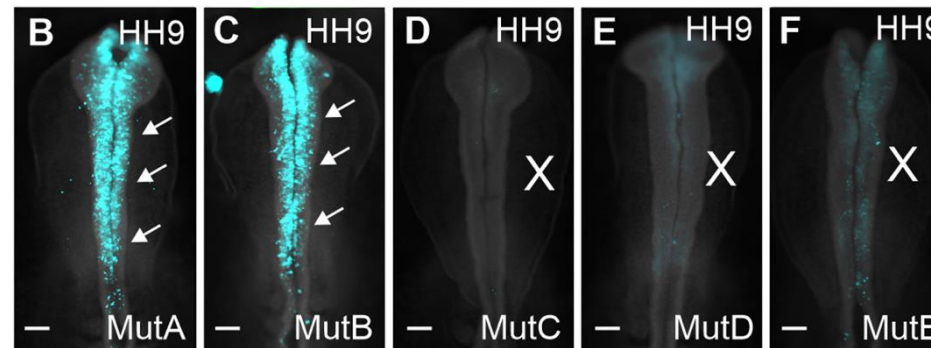
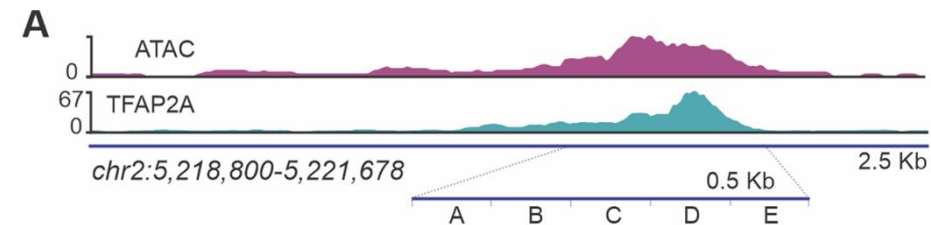
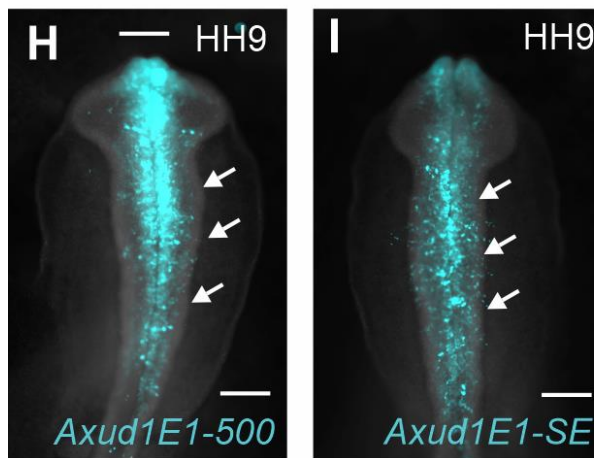
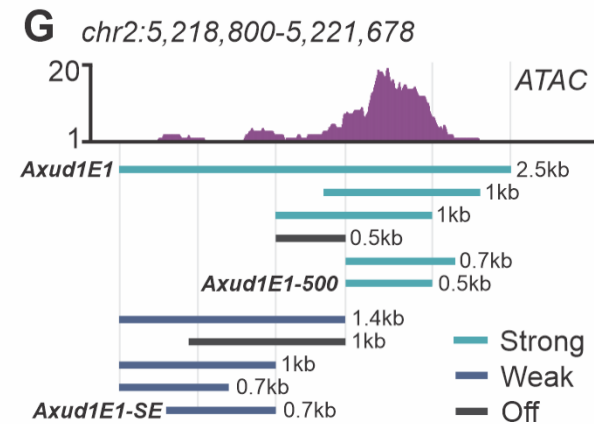


Enhancer-promoter interaction

1. Can we predict enhancer function from epigenetic signatures?
2. How to test and quantify the activity of thousands of candidates (targeted or unbiased)?



Enhancer grammar



M

SP5

CACATTTGTTCAGGGCAGGGGATGGGAGTCTGCTCAGACACAGGTCTCTGCCTT

CGAGGTGGTGGATAAAACGAGGAACAGACTAAAGACGTGCCTTCGGACAAAGG

ATGGGGAGAGCCTGGAGAGAGCGCAGAGCCTGGA

MSX

GCAATGAGTTCTGACCTG

ZIC

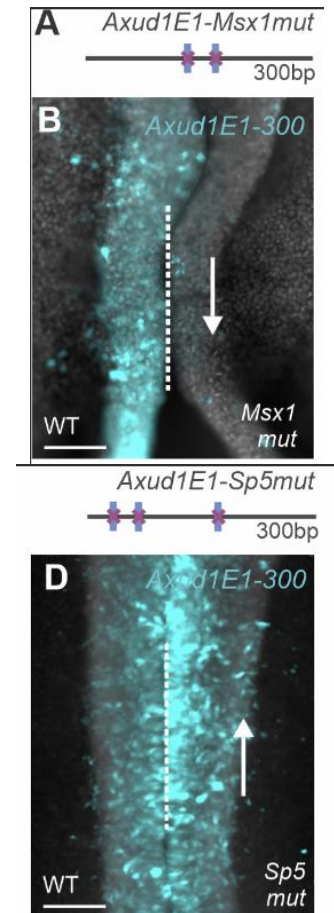
MSX

GCCTGAAAAATGCCAGCACTACAGGGGGCAATAAACCATAGCAGCCCCCGGGG

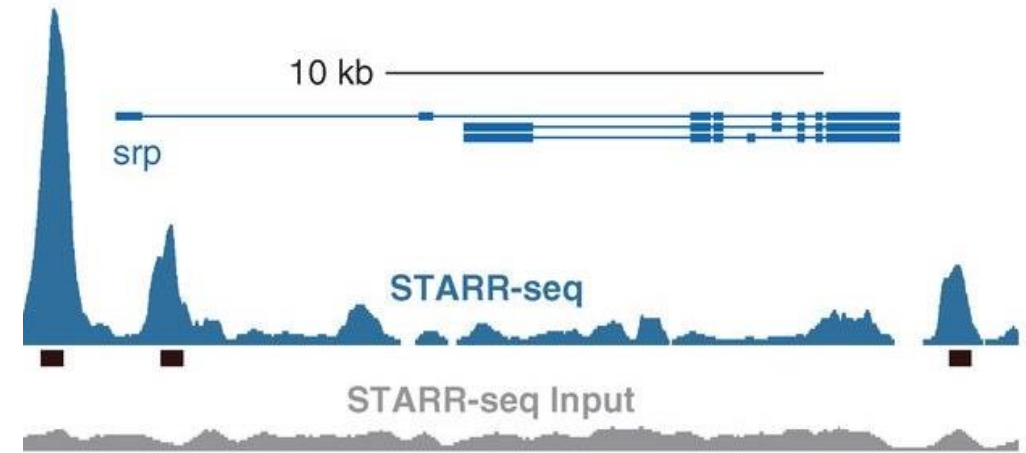
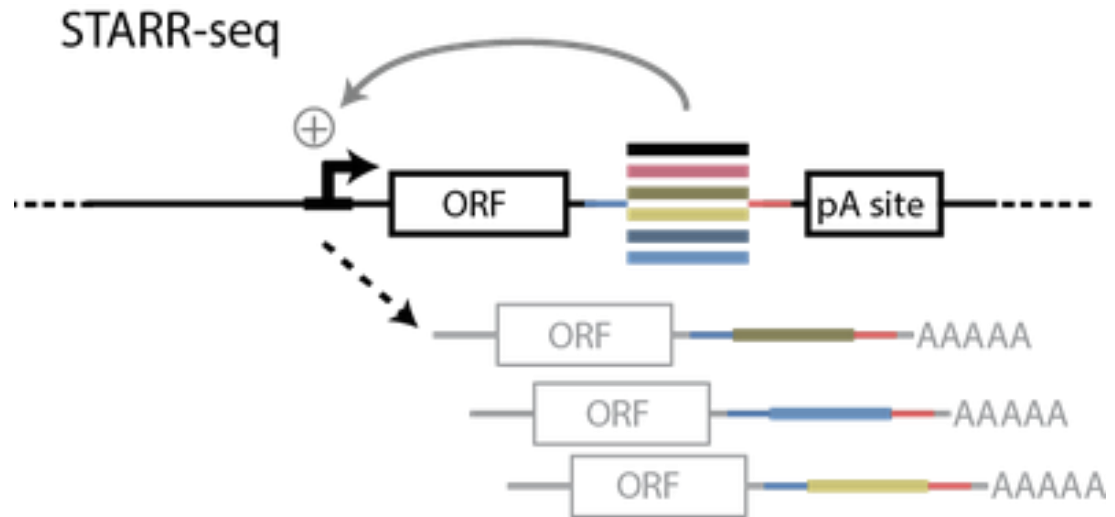
SP5

TGCTTTTAATGAGCTGAATTGCAGTCTTGAGTCCAGCCTTCCATCACTCACAAGC

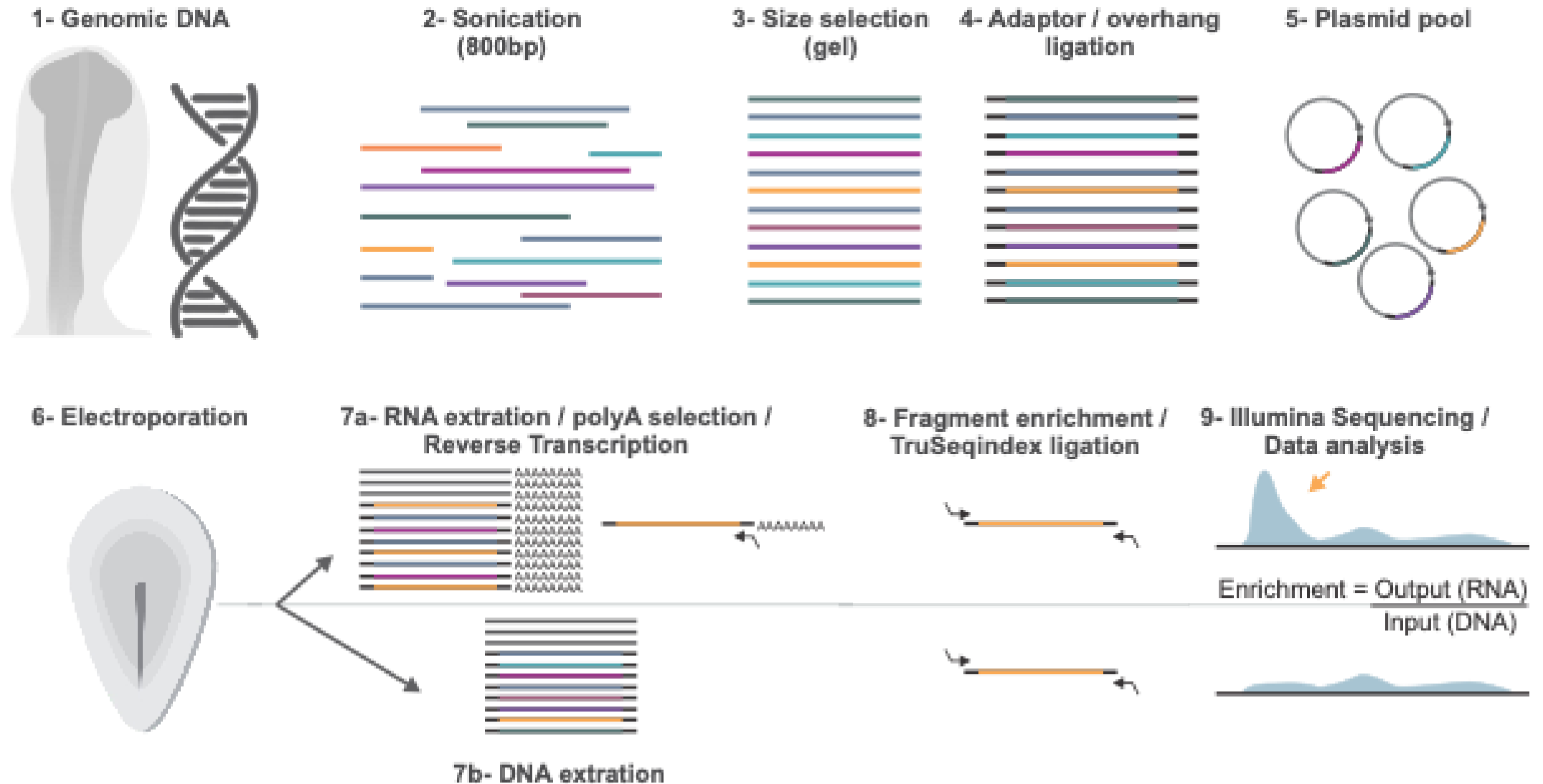
CGCTGCCACAGCTGGTTTTAGCAGATGTCTCCTCCGACAGAAAGGATGCGGCAA



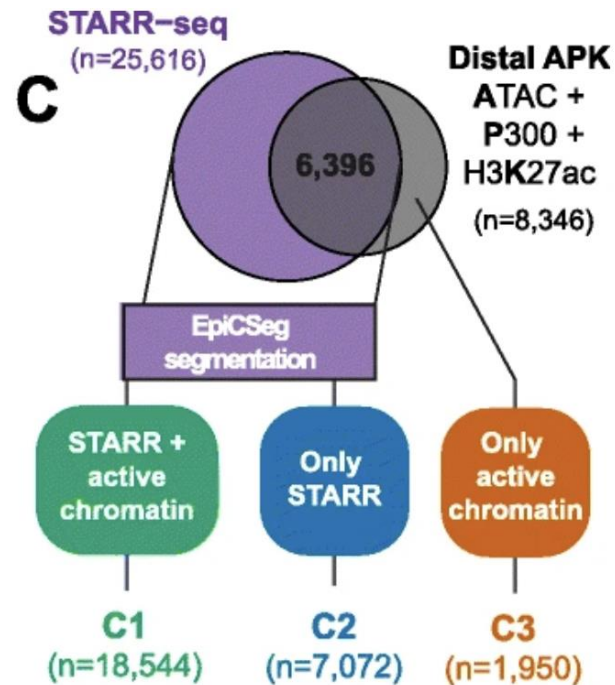
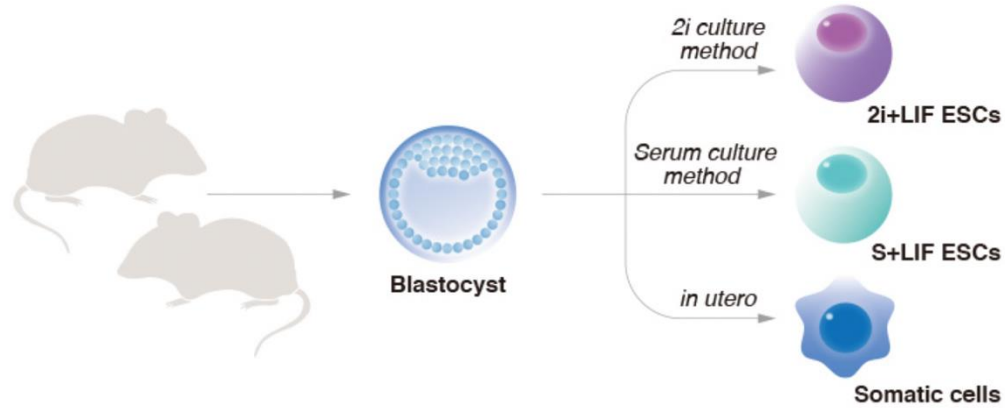
Starr-seq (self-transcribing active regulatory region sequencing)



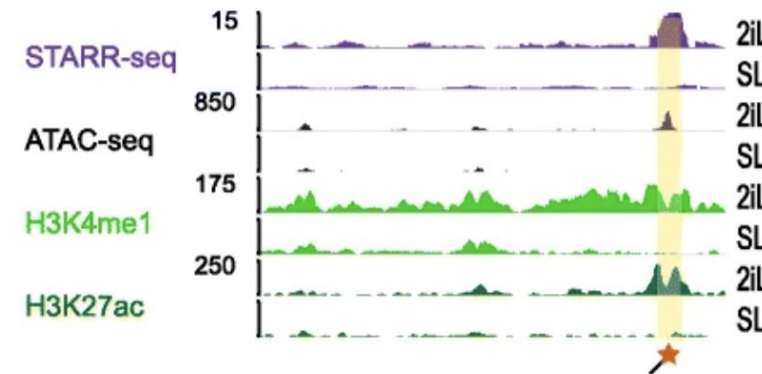
STARR-seq: overview



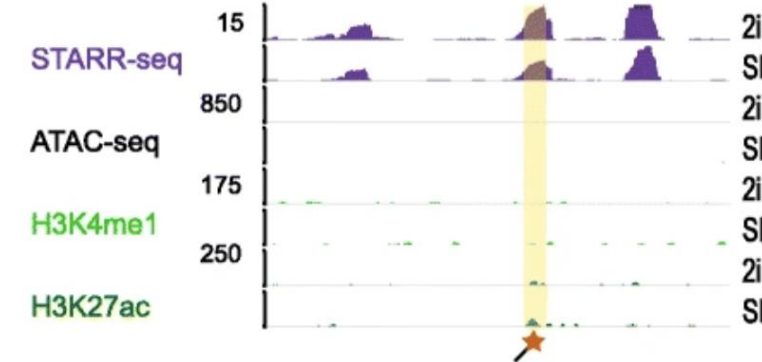
“Closed enhancers” are reactivated during reprogramming



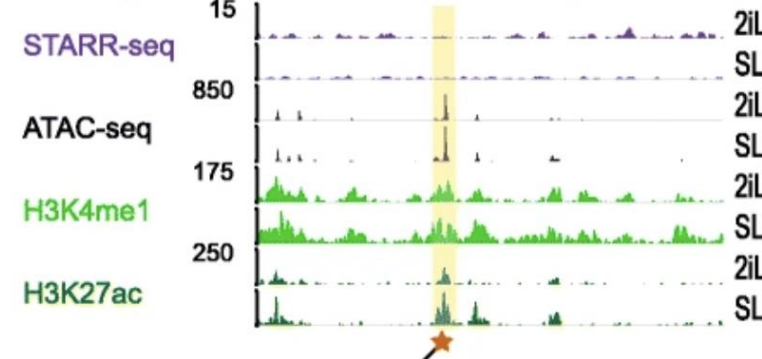
C1, 2iL example chr9: 22,830,000 - 22,853,000 (22 kb)



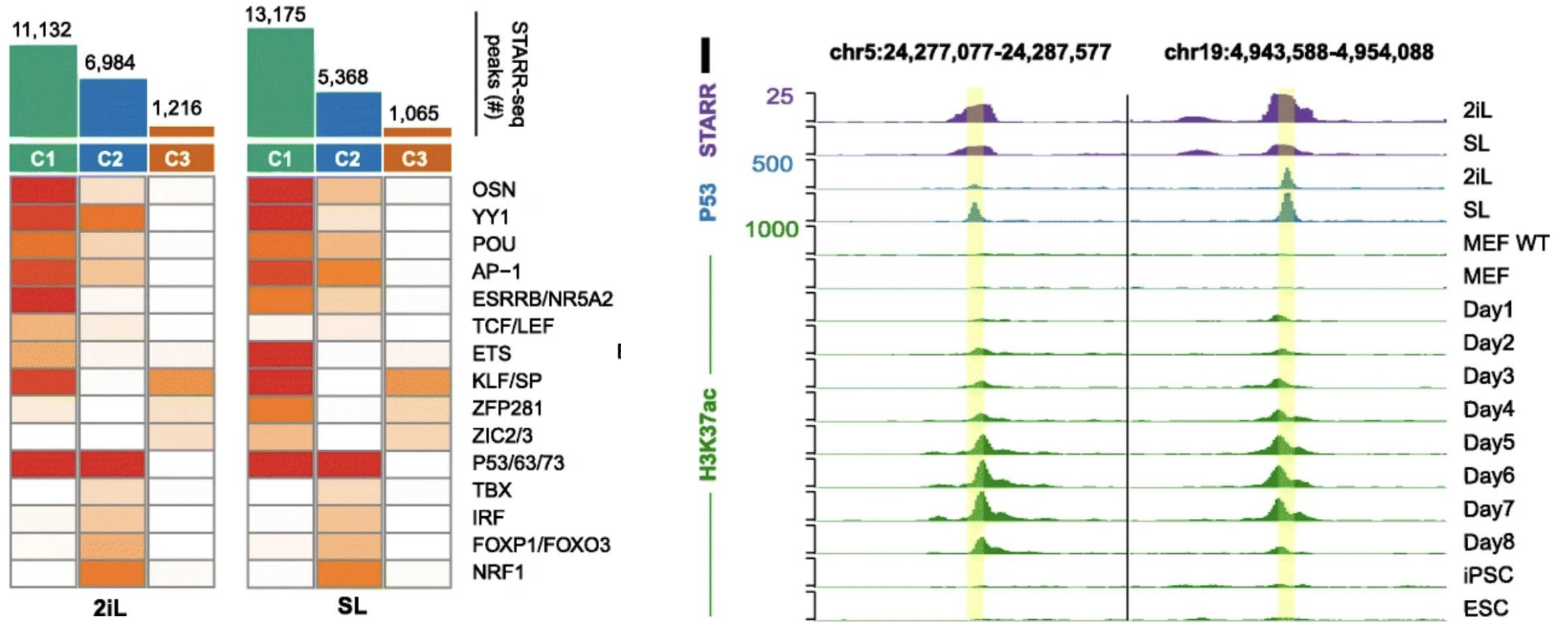
C2 example chr6: 43,865,000 - 43,885,000 (20 kb)



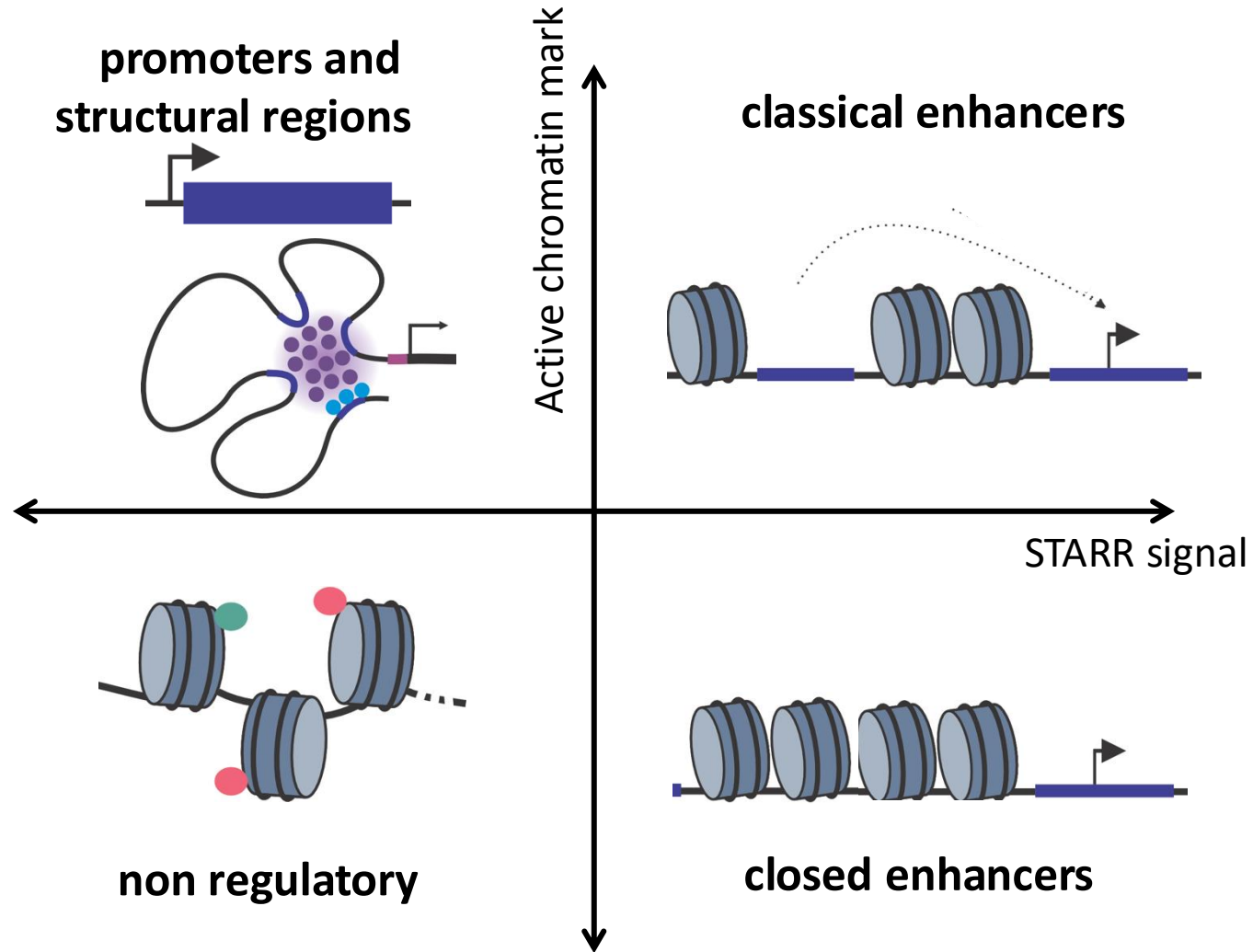
C3 example chr2: 91,822,000 - 91,886,000 (64 kb)



“Closed enhancers” are reactivated during reprogramming



STARR-seq across different chromatin topologies

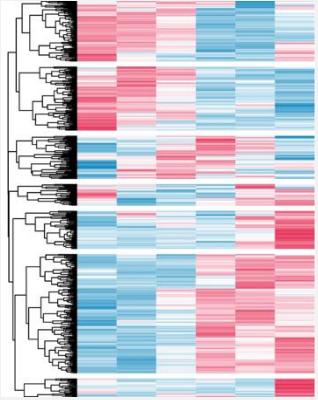


Techniques

GENE EXPRESSION

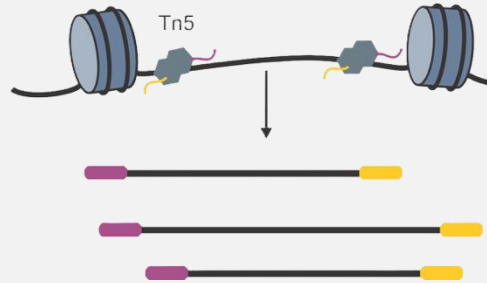
RNAseq

Spatial transcriptomics



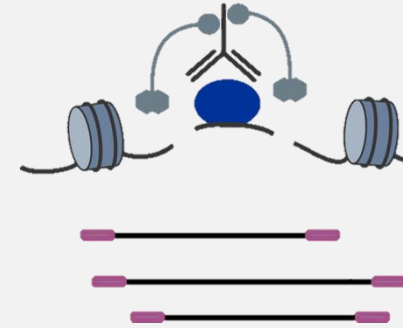
EPIGENOMIC REGULATION

ATACseq



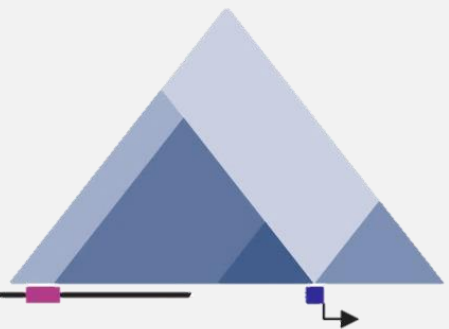
DNA-PROTEIN BINDING

ChIPseq
CUT&RUN
CUT&TAG



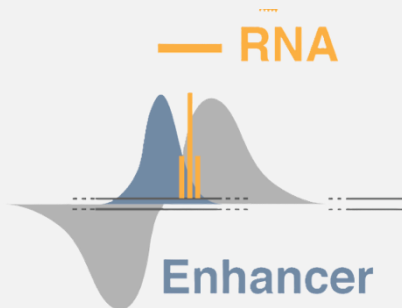
GENOME ORGANIZATION

3D genome organization - HiC



ENHANCER ACTIVITY

eRNA mapping



SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome

